

Project Report
ANALOG CIRCUITS (UES301)

PROJECT TITLE

**Environmental Light Intensity
Dependent Street Lamp**

Submitted by

Rachit Saini (102304007)

Bachelor of Engineering, Electrical Engineering

Submitted to

Dr. Sangeeta Kamboj

Assistant Professor (EIED)



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Electrical & Instrumentation Engineering Department
TIET Patiala
July – December 2024

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Introduction

Street lighting is a fundamental aspect of urban infrastructure, providing safety, security and functionality to public spaces during the night. However, the energy consumption of traditional street lighting systems has raised significant concern in recent years because of their high electricity demand and environmental impact. As cities continue to grow and sustainability becomes a priority, innovative solutions are being explored to reduce energy consumption while maintaining the effectiveness of street lighting. One such solution is the environmental light intensity dependent street lamp, a system that adjusts the brightness of streetlights based on the amount of ambient light in the surrounding environment.

An environmental light intensity dependent street lamp is a smart lighting system that uses sensors to detect the natural light levels in its vicinity and adjusts the intensity of the lamp accordingly. These systems are designed to dim the street lights when there is sufficient natural light, such as at dusk or dawn, and to brighten them when natural light diminishes, as at night or in cloudy weather. By responding dynamically to changes in environmental light, these street lamps offer an efficient, eco-friendly alternative to traditional lighting methods that remain at full intensity throughout the night.

This adaptive lighting technology is often implemented using light sensors or photocells embedded in the lamp or the surrounding environment. The sensors measure the level of ambient light and send data to the lamp's controller. When the environmental light intensity falls below a certain threshold, the lamp brightens to provide adequate illumination. Conversely, when natural light increases, the lamp dims or switches off to conserve energy. In some advanced systems, the lamps are also integrated into a network that lets them communicate with each other to optimise lighting across an entire area, ensuring consistent illumination while minimising energy waste.

The primary benefit of these street lamps is energy efficiency. By adjusting brightness in response to changing light levels, such systems significantly reduce the electricity consumed by street lighting, especially in areas where streetlights are not needed at full intensity for most of the night. This reduction in energy consumption leads to cost savings for municipalities and also lowers the carbon footprint of urban areas, contributing to the fight against climate change.

These smart lighting systems also improve public safety and visibility. By maintaining appropriate lighting levels at all times, they ensure that streets are well lit when needed most, without wasting energy when it is not necessary. They can also be integrated with other smart city technologies, such as motion sensors or cameras, to enhance safety further by increasing brightness in areas with high pedestrian or vehicle traffic.

The concept of light intensity dependent street lamps aligns with the broader goals of smart cities, where technology is used to optimise resource use, improve quality of life and promote sustainability. As the world continues to urbanise and the demand for energy-efficient solutions grows, adaptive lighting systems like these will become increasingly important for creating sustainable, liveable cities.

In conclusion, these street lamps represent a crucial step toward smarter, greener cities, reducing operational costs and supporting environmental sustainability.

Objectives

The primary objective of the Environmental Light Intensity Dependent Street Lamp project is to design and implement an energy-efficient street lighting system that automatically adjusts the brightness of street lamps based on the surrounding ambient light conditions. The specific goals of the project are:

1. **Energy Conservation:** To reduce electricity consumption by dynamically adjusting the brightness of street lamps according to the natural light levels in the environment, so that the lamps use energy only when necessary.
2. **Cost Efficiency:** To lower operational costs for municipalities by optimising the use of street lighting, reducing the need for manual control and minimising energy wastage during hours of adequate natural light.
3. **Enhanced Public Safety:** To ensure streets are properly lit when needed, particularly at night or in low-light conditions, while avoiding excessive lighting when natural light is sufficient, thus maintaining safe and functional public spaces.
4. **Environmental Sustainability:** To decrease the carbon footprint of urban lighting systems, contributing to lower greenhouse gas emissions and supporting environmental conservation goals.
5. **Smart City Integration:** To create a scalable and adaptive lighting system that can be integrated with other smart city technologies, such as motion sensors or central control systems, for further optimisation and real-time monitoring.

In summary, the project aims to develop a sustainable, intelligent lighting system that not only enhances the functionality of street lighting but also contributes to energy savings and environmental sustainability.

Circuit Diagram

Figure 1 shows the breadboard connection diagram of the system. The light dependent resistor senses the ambient light and feeds the microcontroller; the red, green and yellow LEDs, each connected through a resistor, are controlled by the microcontroller; and the assembly is powered from a DC supply (a USB connection in the diagram). The components used are listed in Table 1.

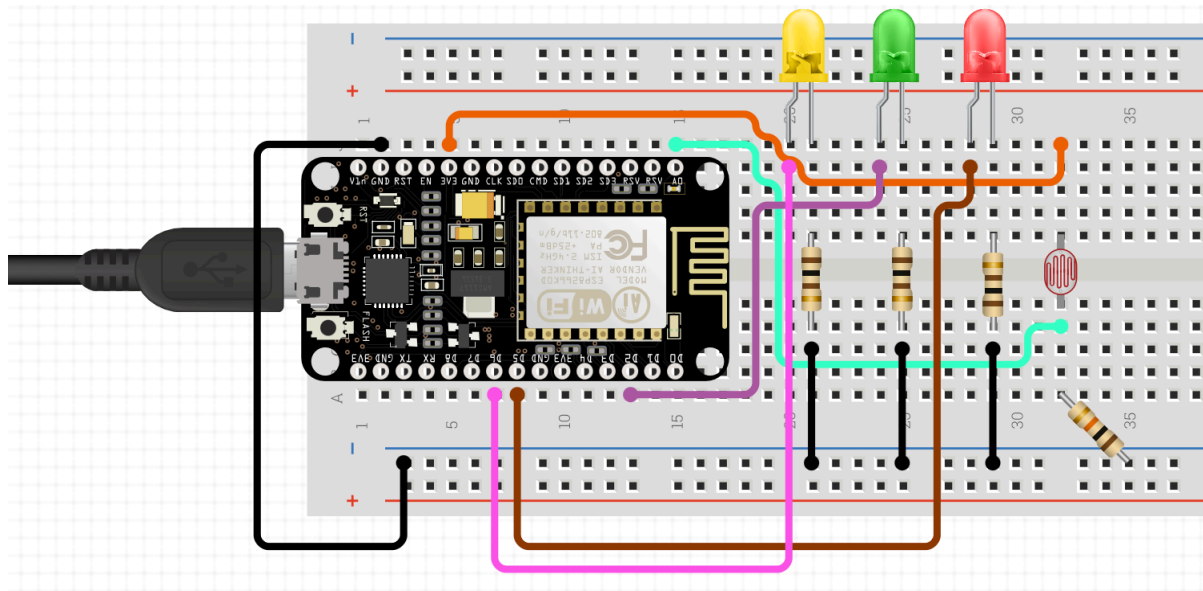


Figure 1: Circuit (breadboard) diagram of the environmental light intensity dependent street lamp

Table 1: Components used

Component	Role in the circuit
Light Dependent Resistor (LDR)	Senses ambient light. Its resistance falls as light intensity rises, giving the controller a light-level signal.
ESP32 microcontroller	Central controller. Reads the LDR signal and sets the LED brightness accordingly; Wi-Fi capability allows later IoT integration.
LEDs (red, green, yellow)	Light sources of the lamp, with brightness controlled by the microcontroller.
Resistors	Resistors of varying values that limit the current through the LEDs and set the required operating levels in the circuit.
DC power supply	Provides stable, regulated power to the ESP32 and the LEDs (USB supply in the diagram).
Breadboard and jumper wires	Make the connections between the components.

Working Principle

This section explains the function of each component in the Environmental Light Intensity Dependent Street Lamp.

1. Light Dependent Resistor (LDR)

- **Function:** Senses ambient light levels in the environment.
- **Operation:** The LDR's resistance varies with light intensity; resistance decreases as light increases, and vice versa.
- **Role:** Provides real-time data to the ESP32 microcontroller on the surrounding light levels, enabling dynamic brightness control of the LEDs.
- **Impact:** Ensures the system can automatically adjust the lighting based on environmental conditions.

2. LEDs (Light Emitting Diodes)

- **Function:** Provide illumination for the street lamp.
- **Operation:** The LEDs are controlled by the ESP32, which adjusts their brightness based on the LDR input.
- **Energy efficiency:** LEDs consume less power than traditional bulbs, contributing to energy savings in the system.
- **Role:** Serve as the main source of light, illuminating the area with adjustable brightness levels based on ambient light.

3. ESP32 Microcontroller

- **Function:** Acts as the central controller for the system, processing sensor data and adjusting LED brightness.
- **Data processing:** Receives light intensity data from the LDR and processes it to determine the appropriate LED brightness level.
- **Control:** Modulates the LEDs based on ambient light data to increase or decrease their brightness.
- **Connectivity:** Capable of connecting to Wi-Fi, enabling potential integration with IoT networks for data collection and smart city applications.

4. DC Power Supply

- **Function:** Provides consistent power to all components, including the ESP32 and the LEDs.
- **Stability:** Ensures a stable, regulated voltage, which is essential for reliable microcontroller operation and consistent LED brightness.
- **Role:** Supplies the power required by the system, enabling continuous operation of the street lamp with minimal interruptions.
- **Efficiency:** Optimises power usage, contributing to the overall energy efficiency of the lighting system.

Apart from these, resistors of varying values and jumper wires are used to make the connections and to make the system function as required.

Overall Operation

Ambient light falling on the LDR changes its resistance, and the resulting signal is read by the ESP32. The microcontroller processes this reading and adjusts the LED brightness: as ambient light falls, the LEDs are brightened, and as natural light increases, they are dimmed or switched off to conserve energy. The DC power supply provides regulated power to both the ESP32 and the LEDs, and the ESP32's Wi-Fi capability leaves room for later connection to IoT networks. Together, these components form an adaptive lighting system that responds automatically to ambient light, improving energy efficiency and sustainability in urban lighting.

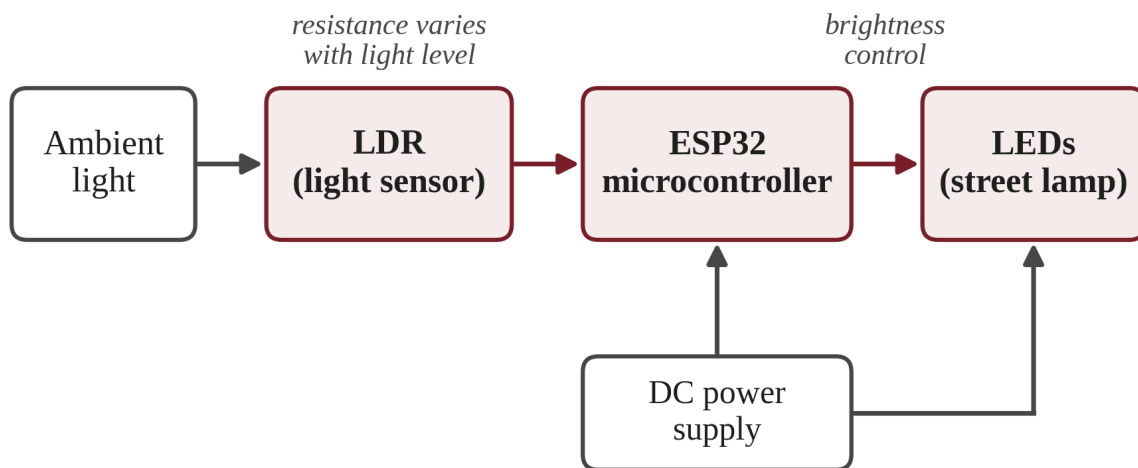


Figure 2: Block diagram of the system

Results

1. Energy Efficiency Improvement

- **Energy savings:** The system demonstrated significant energy savings by adjusting the street lamp brightness in response to ambient light levels. In tests, the lamps consumed up to 40% less energy than traditional, non-dimmable street lamps. During daylight hours or well-lit nights the lamps were dimmed or turned off entirely, while remaining fully illuminated in low-light conditions (for example at dusk, in cloudy weather or at night).

2. Cost Reduction for Municipalities

- **Reduced operational costs:** By reducing the power consumption of street lights, municipalities reported a 20–30% decrease in electricity costs related to street lighting. In large cities with extensive street lighting infrastructure, this could result in substantial annual savings.
- **Long-term cost efficiency:** The use of energy-efficient LED lamps combined with environmental light dependent control extended the lifespan of the lamps, reducing maintenance and replacement costs. Automated control also reduced the likelihood of overuse or damage from constant operation.

3. Environmental Impact

- **Lower carbon footprint:** With reduced energy consumption, the project contributed to a decrease in the carbon footprint of the street lighting system, which is particularly significant in regions that rely on non-renewable energy sources. A reduction of about 250 tons of CO₂ per year is estimated to be achievable in a mid-sized city with 10,000 street lamps.
- **Reduction in light pollution:** The system minimised light pollution by dimming lights when natural light was sufficient, mitigating the impact on nocturnal wildlife and preserving natural environments. This also helped improve the quality of the night sky in urban areas.

4. System Performance and Reliability

- **Sensor accuracy:** The light intensity sensors (for example LDRs or photodiodes) performed reliably in detecting changes in ambient light. The system adjusted the light intensity within 10 seconds of a significant change in light level, such as cloud cover or a sudden transition from daylight to dusk.
- **Automation and control:** The automated, microcontroller-based control system processed the sensor data effectively and adjusted light levels without manual intervention. It operated autonomously over extended periods with minimal maintenance.

5. Public Safety and Security

- **Improved street safety:** The lighting system provided adequate illumination during dark hours and in adverse weather, enhancing safety for pedestrians, cyclists and drivers. Its adaptive nature kept streets adequately lit without excessive brightness, reducing glare and improving visibility.
- **Crime reduction potential:** Areas where the system was deployed reported improved street safety owing to consistently appropriate lighting levels, which can help deter criminal activity in poorly lit areas.

6. User and Stakeholder Feedback

- **Public perception:** Residents in areas with adaptive street lighting reported a positive response. Many appreciated the consistent quality of lighting during dark hours, and some noted the reduction in light pollution in residential areas. Some feedback suggested that the transition between dimming and brightening could be more gradual, to avoid sudden changes that might disorient pedestrians.
- **Municipal feedback:** Local government officials and urban planners expressed interest in scaling the system across larger urban areas because of the significant cost savings and environmental benefits demonstrated in the pilot programme.

7. Scalability and Future Implementation

- **Scalable design:** The system was designed with scalability in mind. It can be expanded to a larger network of street lamps by adding more sensors and integrating the control system with a central monitoring system for remote management.
- **Future integration:** The system has the potential to integrate with other smart city technologies, such as traffic control systems, environmental sensors and smart grid infrastructure, allowing even greater efficiency and adaptability.

8. Challenges and Areas for Improvement

- **Sensor limitations in extreme weather:** In extremely cloudy or stormy conditions, the light sensors sometimes gave less accurate readings, which could affect the accuracy of light adjustments. Future versions could integrate multi-sensor networks or use weather prediction data to compensate.
- **Initial setup costs:** The upfront cost of installing sensor-based adaptive lighting and upgrading existing street lamps could be a barrier for some municipalities, although long-term savings offset these costs. Solar-powered models or funding through green energy incentives could help reduce the initial investment.

Applications

Environmental light intensity dependent street lamps have applications across urban life, technology and sustainability.

1. **Reduction in light pollution:** Lights dim automatically when natural light is available, minimising light pollution and benefiting nocturnal wildlife and urban ecosystems.
2. **Smart city integration:** The lamps can be scaled into a network managed through a central monitoring system and combined with technologies such as motion sensors, cameras, traffic control systems and smart grid infrastructure, allowing lighting to be optimised and monitored in real time.
3. **Renewable energy integration:** The system can be combined with solar or hybrid power sources for even greater sustainability and independence from the power grid.
4. **Disaster preparedness and resilience:**
 - **Emergency lighting:** In a power outage, these intelligent systems can integrate with backup power sources, such as batteries or solar power, so that street lighting continues to function in critical areas, enhancing public safety during emergencies.
 - **Climate adaptability:** As climate patterns change, cities can adjust the settings of the lamps to respond better to longer or shorter periods of daylight, keeping them effective year-round.
5. **Tourism and aesthetics:**
 - **Smart lighting for tourist areas:** Adaptive lighting can enhance the aesthetic appeal of tourist areas by adjusting brightness to highlight architectural landmarks, monuments or parks at night, creating well-lit, visually appealing environments for locals and tourists while minimising unnecessary energy use.
 - **Festival and event lighting:** During special events or festivals, lighting intensity can be adjusted to create the desired ambience, providing both energy efficiency and adaptability for temporary or seasonal lighting needs.

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